See attached instructions.

https://gemini.google.com/app/a8f5376e084a5059

User prompt: See attached instructions.

Response: Adversarial Review: Study 1 (Positive Control), Round 5 — Part 1

1. Wall 3 on even parts (D9)

- **agreed:** Negating target positions negates linear drift capture while preserving symmetric fee and ruin drags, and two programs with strictly opposite effects (`L2.8`) never both win 39 of 57 pairs.
- **disputed:**
  - **Where:** `architecture.md` 5.1 ("Negating every position negates the first, the drift captured. It leaves the second: the variance path, liquidation, clipping, ruin. … Wall 3 measures the second part directly, on mirror pairs of programs.") and 5.5 / `briefs/1-positive-control.md` Step 5 (`BLIND_MARKETS = 200`, `EVEN_ALLOWED = 9` over 1,000 pooled classifications across 5 mirror pairs).
  - **The construction, with numbers:**
    1. **The variance path and bar-return clipping are odd in position, not even.** For a DSL program p with target weights $w_{i,t}$ and its mirror $-p$ with weights $-w_{i,t}$, absent margin liquidation, the bar-t return is $r_{p,t} = g_t - c_t$ and $r_{-p,t} = -g_t - c_t$, where $g_t = \sum_i w_{i,t} r^{\text{net}}_{i,t}$ is the gross return (including `generate`'s bar-return clip) and $c_t \geq 0$ is the turnover fee. Because `w_t` is chosen before bar t's return is realized, $\text{Cov}(g_t, c_t) \approx 0$ and the return standard deviation $\sigma_p(m)$ (which carries N2's GARCH variance path) is symmetric: $\sigma_{-p}(m) = \sigma_p(m)$. Therefore, dividing the odd gross mean $\bar{g}_p(m)$ by the even standard deviation $\sigma_p(m)$ yields an **odd** term that cancels identically in `even(p, -p)`:

       $$\frac{S(p,m) + S(-p,m)}{2} = \quad 2190\,\frac{(\bar{g}_p(m) - \bar{c}_p(m)) + (-\bar{g}_p(m) - \bar{c}_p(m))}{2\,\sigma_p(m)} = -\quad 2190\,\frac{\bar{c}_p(m)}{\sigma_p(m)} \leq 0.$$

       Thus, when N2's variance path raises volatility on the planted market ($\sigma_p(m) > \sigma_p(m_0)$), any drift-blind program with negative sample mean $\bar{g}_p < 0$ gets a positive Sharpe effect $\Delta(p) = |\bar{g}_p| \left(\frac{1}{\sigma_p(m_0)} - \frac{1}{\sigma_p(m)} \right) > 0$, while its mirror gets $\Delta(-p) = -\Delta(p) < 0$; their average $\Delta_{\text{even}}(p, -p)$ is 0. Wall 3 is completely blind to N2 variance-path and return-clipping effects on the Sharpe denominator.

    2. **800 of the 1,000 cells in Wall 3 have zero turnover after bar 0, diluting the threshold by $5\times$.** Four of the five mirror pairs in Step 4 (constant long/short and the three fixed random leverage pairs) hold static weights $w_{i,t} \equiv w_i$. Unless they liquidate, their turnover after bar 0 is zero, so $\bar{c}_p(m) = \bar{c}_p(m_0) = c_0/T \approx 10^{-6}$ and $\Delta_{\text{even}} \approx 0$ on every pair ($q = 0$, not $q = 1/2$). Pooling 800 structurally zero cells with 200 momentum/reversal cells under `EVEN_ALLOWED = 9` lets the single active pair earn on $9/200 = 4.5\%$ of markets—$11.7\times$ the single-pair bound of 0.00384—without tripping Wall 3.
  - **Outcome it changes:** Replaces the pooled `9 of 1,000` check with a per-mirror-pair bound of at most `3 of 200` (`binom.sf(3, 200, 0.003837) = 0.0076`) or replaces the static random pairs with dynamic turnover pairs, and corrects 5.1's claim that `even` measures the N2 variance path.
  - **Severity:** Medium.

2. The void rule (D10)

- **agreed:** The frozen harness cannot resume a crashed session, and judging a cut-off session on a partial ledger penalizes an outage (A24).
- **disputed:**
  - **Where:** architecture.md 5.2 ("The rule reads how the session ended and never a score. The session counted for a market is the first to reach a designed end, markets stay independent, and the null is about such sessions. … The lean is second order while voids are few") and briefs/1-positive-control.md Step 8 Behaviour 2 and 6.
  - **The construction, with numbers:** A session's duration is endogenous to its search success. When the agent finds a high-Sharpe program early (e.g., within 10 evaluations / ~12 turns), it exits cleanly (exit == 0). When it fails to find a strong program on a market, it continues searching across many more turns and tool calls, increasing its exposure to API timeouts, harness crashes, or runner kills (crashed(run) == True). Suppose on a market where the single-session null holds ($p_A = p_{\mathrm{G100}} = 0.15$), a winning session (short) crashes with probability $c_W = 0.02$, while a losing session (long) crashes with probability $c_L = 0.25$. Because a void triggers up to 3 fresh retries on the same market while G100 never retries, the agent's win probability on the counted session rises to:

    $$p_A^{\mathrm{counted}} \approx \frac{p_A(1-c_W)}{p_A(1-c_W)+(1-p_A)(1-c_L)} = \frac{0.15 \times 0.98}{0.15 \times 0.98 + 0.85 \times 0.75} = 0.1874 > 0.15.$$

    For independent arms at $p_{\mathrm{G100}} = 0.15$, this shifts θ from 0.500 to 0.567 and inflates the sequential test's Type I error from 0.020 to 0.114 ($2.7\times$ the 0.043 bound). Furthermore, Step 8 Behaviour 6 ("Killed during a session: on restart that session is a void… and the market is run again") runs *after* searcher.jsonl is already on disk (Step 7.4), so any cluster preemption or manual restart during a long session grants a fresh draw.
  - **Outcome it changes:** CS1.1 and Step 8 must cap total study-wide void reruns (e.g., at most 5 void reruns across the study before pausing for review) or require that a claim holds under both the primary rule and Step 9.4's first-session recomputation when voided sessions had already spent ≥ 10 evaluations.
  - **Severity:** Medium.

3. Constant positions (D9)

- **agreed:** Leaving the frozen generator unchanged and redefining the claim in 5.1 as earning from the plant by any route (funding, momentum, or constant position), with routes reported, resolves Round 4 Finding 3 without modifying gate/markets.py (D9 conceded on the claim change).
- **disputed:**
  - **Where:** briefs/1-positive-control.md Step 4 ("**The reference** holds, on every coin and every bar, a position in the direction of the plant's signal…"), Step 1 (PLANTED: coins 0 to 5 of 12), and architecture.md 5.5 ("**Reported with the walls, never judged:** each mirror-set program's own rate of earning at the study's strength").
  - **The construction, with numbers:**
    1. **The reference trades all 12 coins while only 6 are planted.** On coins 6 to 11, the reference takes full-size funding positions that earn zero planted drift while adding 6 coins of unplanted return variance $\sum_{i=6}^{11} \sigma^2 s_{i,t}^2$. This doubles the reference's portfolio variance and dilutes its Sharpe effect $\Delta(\mathrm{ref})$ by $1/\sqrt{2} = 0.7071$ relative to the 6 planted coins. Consequently, the pair-win threshold SHARE * Δ(ref) = 0.5 * Δ(ref) requires capturing only $0.5/\sqrt{2} = 35.4\%$ of the 6-coin plant's Sharpe. At $\mathrm{SNR} = 0.12$, a momentum average (decay = 0.95, correlation 0.43 in arithmetic.py.txt) concentrated on the 6 planted coins achieves expected effect $\sqrt{2} \times 0.43\,\Delta(\mathrm{ref}) = 0.608\,\Delta(\mathrm{ref}) > 0.5\,\Delta(\mathrm{ref})$, and a constant position on the 6 planted coins clears $0.5\,\Delta(\mathrm{ref})$ whenever $|\mathbb{E}[s]|/\sqrt{\mathbb{E}[s^2]} \geq 0.354$.

2. **No wall stops calibration if constant positions earn.** Because the mirror-set earning rates in Step 5.5 are "reported, never judged," calibration proceeds automatically into Step 7 even if constant short/long earns on $> 50\%$ of `blind` markets at the calibrated strength b^*.

- **Outcome it changes:** Either restrict the reference program to `PLANTED` (coins 0 to 5) or document that `SHARE = 0.5` against the 12-coin reference equals 0.354 of the 6-coin oracle; and make it a wall if constant long or constant short earns on more than 50% of the 200 `blind` markets.
- **Severity:** Low-Medium.

4. The measure (Step 2)

- **agreed:** Scoring pick and reference on each market's own 57 fresh twin pairs removes selection bias and between-market dependence (`A18`), and the pure/kernel split in `measure.py` is clean.
- **disputed:**
  - **Where:** `briefs/1-positive-control.md` Step 2 ("Laws of the pure layer, for every pair of lists (finite scores drawn as dyadic numbers, so that sums are exact): … **L2.2** Adding one number to both scores of a pair, for either program, changes nothing. … **L2.7** A pick whose effect on every pair is `c` times the reference's wins exactly the pairs where the reference's effect is positive, when `c ≥ SHARE`, and none when `c < SHARE`.") and Behaviour 7 (`scores(program, pairs)`).
  - **The construction, with numbers:**
    1. **IEEE-754 float absorption breaks `L2.2`, `L2.7`, and `L2.9` on unrestricted dyadic numbers.** In Python `float` (53-bit significand), drawing arbitrary dyadic numbers in `hypothesis` without a shared exponent bound causes rounding when adding or subtracting across scales. For example, with dyadic scores `ref_planted = 1.0 + 3 * 2**-52`, `ref_twin = 1.0` (`ref_effect = 3 * 2**-52`), `pick_twin = 1.0`, and `c = SHARE = 0.5`, setting `pick_planted = pick_twin + c * ref_effect = 1.0 + 1.5 * 2**-52` rounds to `1.0 + 2**-52` (or for `ref_twin = 0.5`, rounds down), making `pick_planted - pick_twin < 0.5 * ref_effect` so `L2.7` fails at `c = 0.5`. Similarly, adding `x = 2**30` in `L2.2` zeroes out differences of 2^{-25}.
    2. **Streaming signature in Behaviour 7 (`scores(program, pairs)`).** `gate.search.Evaluator` streams multiple *programs* against *one* loaded market directory (`mm eval`), not multiple markets against one program. Calling `scores(program, pairs)` with a single `program` across 57 pairs (114 distinct markets) can evaluate only 1 program per `Evaluator` process on each fresh market.
  - **Outcome it changes:** In Step 2's test specification, constrain dyadic test inputs to fixed-point integers $k \cdot 2^{-16}$ with $|k| \leq 2^{20}$ so sums and multiplication by 0.5 are exact in `float64`; and change Behaviour 7 to `scores(programs, pairs)` (accepting a sequence of programs per market) so `[pick, reference]` in Step 5 and the 11 programs in Wall 3 stream through one `Evaluator` per fresh market.
  - **Severity:** Medium (build blocker in Step 2 tests and Step 5 runtime).

5. The 39-of-57 rule and its bound

- **agreed:** Because each fresh pair is seeded independently by its id (`L1.4`) and `generate` draws fresh innovations and funding per id, pair wins are conditionally $\mathrm{Binomial}(57, q)$ given the pick; $(57, 39)$ is the exact minimum rule satisfying $\mathbb{P}(\mathrm{win} \geq 39 \mid q = 1/2) = 0.003837 \leq 0.005$ and $\mathbb{P}(\mathrm{win} \geq 39 \mid q = 3/4) = 0.9012 \geq 0.90$, and `L2.10` holds.
- **disputed:** None.

6. Calibration (Step 5)

- **agreed:** Sizing the search at 1,600 markets (`TOLERANCE = 0.005`) and the check at 10,000 markets (`BAND = 0.12 to 0.18`) reduces the false-stop probability from 31% to 0.0046 (`A24`).
- **disputed:**
  - **Where:** `briefs/1-positive-control.md` Step 5 Behaviour 1 ("For a rising function that is below `TARGET - TOLERANCE` at strength zero, above `TARGET` at the upper end, and rises by at most 0.01 over any 0.01/4096 of strength, the search returns a strength.") and Behaviour 6 ("Before the first kernel run the job times 20 evaluations and prints the projected time… 328 evaluations a market at one strength, at most 13 strengths on 1,600 markets and one on 10,000, 10,102,400, plus wall 3's 250,800. A projection above 24 hours stops the job: a wall.").
  - **The construction, with numbers:**
    1. **Boundary equality in Behaviour 1.** The tolerance band `[0.145, 0.155]` has width $2 \times 0.005 = 0.01$. After `MAX_STEPS = 12` midpoints on `[0, 0.01]`, the bracket width is $0.01/2^{12} = 0.01/4096$. A rising function with step 0.01 over $0.01/4096$ can take the value 0.145 at the lower endpoint and 0.155 at the upper endpoint; if the builder writes strict inequality `abs(rate - TARGET) < TOLERANCE` instead of $\leqslant$ `TOLERANCE`, both endpoints fail and Behaviour 1's property test breaks.
    2. **The 20-evaluation timing in Behaviour 6 misses market generation and process startup, and recomputes `cal-a` twins 13 times.** Timing 20 evaluations on one market measures warm streaming (`9 ms/eval`), omitting `generate("N2", ...)` + `write_pair` and per-market `Evaluator` startup across up to $13 \times 1,600 \times 57 + 10,000 \times 57 = 1,755,600$ pairs ($3,511,200$ 730-day markets). Moreover, on `cal-a`, the $1,600 \times 57 = 91,200$ twin markets have fixed strength 0 across all 13 bisection steps, and the reference's twin score is identical at every step; regenerating and re-scoring them 13 times wastes $12 \times 1,600 \times 57 \times 2 = 2,188,800$ evaluations and $1,094,400$ market writes.
  - **Outcome it changes:** Specify `abs(rate - TARGET)` $\leqslant$ `TOLERANCE` in Step 5; cache `cal-a`'s strength-0 twins and reference twin scores across bisection steps; and time a full calibration market (including `write_pair` and `Evaluator` startup) in Behaviour 6.
  - **Severity:** Medium (operational / 24-hour wall risk).

7. The sequential test ($\theta_1 = 0.7$, at most 100 markets)

- **agreed:** Exact integer characterization `L6.1` ($7^w 3^l \geq 19 \cdot 5^{w+l}$ and $19 \cdot 7^w 3^l \leq 5^{w+l}$, with no boundary ties since $\gcd(21, 95) = 1$), dominance `L6.4`, concordant invariance `L6.5`, worst-case size $0.0428 \leq 0.043$ `L6.6`, power rows across `[0.11, 0.19]`, and the 2-week account-window table in 5.3 all verify by exact enumeration.
- **disputed:** None.

8. The criterion draft CS1.1

- **agreed:** `CS1.1` accurately states the population, the paired sign test, the 0.043 size bound, the non-rejection naming rule (never "no better"), and the underpowered region at $p_A = 0.3$.
- **disputed:**
  - **Where:** `briefs/1-positive-control.md` Step 0 (`CS1.1`, "*Passes while the claim is false:* only by chance, at most 0.043, or by a measure that pays something other than the drift captured. Wall 3 bounds the second before any session.").
  - **The construction, with numbers:** As shown in Points 1 and 2 above, `CS1.1`'s "*Passes while the claim is false*" clause omits (a) endogenous crash-length selection bias under the void rule (which raises Type I error from 0.020 to 0.114 when $c_L = 0.25, c_W = 0.02$ at $p = 0.15$) and (b) the fact that Wall 3's `even` cancels odd variance-path effects and is diluted $5\times$ by pooling 800 static-position cells with 200 dynamic cells.

- **Outcome it changes:** Add one clause to `CS1.1`'s "*Passes while the claim is false*" bullet noting that differential crash rates between winning and losing searches can bias reruns (bounded by reporting Step 9.4's first-session recomputation and capping voids), and enforce Wall 3's threshold per mirror pair (≤ 3 of 200).
- **Severity:** Low-Medium.

Convergence Ledger (After Round 5 on Study 1)

- **Agreed (Closed in Round 5):**
  - **A25 (closes D9 on the claim and generator):** Redefining Study 1's claim to earning from the plant by any route (reporting the inputs read) while keeping the frozen generator unchanged is valid.
  - **A26:** The 39-of-57 pair rule (`L2.10`) has exact Binomial size $0.00384 \leq 0.005$ at $q = 1/2$ and power $0.901 \geq 0.90$ at $q = 3/4$.
  - **A27:** Calibration sizes ($1,600$ search markets at tolerance 0.005; $10,000$ check markets at band $[0.12, 0.18]$) bound false stops at 0.0046.
  - **A28:** The sequential sign test ($\theta_1 = 0.7$, $N_{\max} = 100$), its integer laws `L6.1`–`L6.7`, its power table across $[0.11, 0.19]$, and the two-week window table are exact.
- **Still Open (Study 1 Build & Criterion Adjustments Before Commit):**
  - **D9-a (Wall 3 pooling and odd variance path):** Enforce Wall 3 per mirror pair (≤ 3 of 200 instead of ≤ 9 of 1,000 pooled across four zero-turnover pairs) and note in 5.1 that `even` measures fee/liquidation asymmetry, not the N2 variance path.
  - **D9-b (Constant-position gate & 12-coin reference dilution):** Make constant long/short earning $> 50\%$ on `blind` markets a wall (or restrict the reference to `PLANTED` coins 0–5 so `SHARE = 0.5` is not diluted to 0.354 of the 6-coin plant).
  - **D10 (Endogenous void bias in CS1.1):** Note differential crash probability in `CS1.1`'s "*Passes while the claim is false*" bullet and cap total study voids (or flag if Step 9.4's recomputation flips a `claim`).
  - **Build plan fixes (Steps 2 & 5):** Bound dyadic exponents in Step 2's Hypothesis laws (`L2.2`, `L2.7`, `L2.9`); batch `[pick, reference]` in one `Evaluator` per fresh market in Step 2 Behaviour 7; use `⩽ TOLERANCE` and cache `cal-a` twins ($b = 0$) in Step 5.
